SAT Math Mastery · Session 10

Circles: Equations, Arcs and Tangents

Time	3 hours: Part A in class (about 2 hours of solving), then Part B, the self-assessment, done independently (about 50 minutes), and a short error log.
Topics	circle equations: completing the square, center and radius · writing a circle's equation from conditions · arc length and sector area · chords, diameters and inscribed angles · tangent lines and radii · lines meeting circles · inscribed polygons
Name / Date	_____ _____

PART A · IN CLASS — 35 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal). Grey boxes show exactly what to type in Desmos, one expression per line; tap a grey dot to read an intersection, intercept or vertex. A letter such as a, c or k becomes a slider when you accept Desmos's offer to add one.

1. HARD · DESMOS

In the xy-plane, the graph of $2x^2 - 8x + 2y^2 + 12y = 24$ is a circle. What is the radius of the circle?

(A) $\sqrt{50}$ **(B) 5 ✓**

(C) 6.5 (D) $\sqrt{40}$

1. Type the circle exactly as given: `2x^2-8x+2y^2+12y=24`. Desmos draws it.
2. Center: for $2x^2 - 8x$ the center's x is $\frac{8}{2 \cdot 2}$: Type `8/(2*2)`: it returns 2; for $2y^2 + 12y$ it is $-\frac{12}{2 \cdot 2}$: Type `-12/(2*2)`: it returns -3.
3. Draw the horizontal line through the center, `y=-3`, and tap where it meets the circle: $(-3, -3)$ and $(7, -3)$.
4. Type `(7-(-3))/2`: it returns 5, the radius, choice (B).

Answer: (B)

2. HARD · DESMOS

A circular roundabout has its center at $(2, -5)$ in a coordinate model of the area, where distances are measured in hundreds of meters. Road k is tangent to the roundabout at the point $(9, -1)$. What is the slope of road k?

1. Type the center `(2,-5)` and the point of tangency `(9,-1)`.
2. Slope of the radius: Type `(-1-(-5))/(9-2)`: it returns 0.5714 (that is $\frac{4}{7}$).
3. The tangent is perpendicular to the radius, so use the negative reciprocal: Type `-1/((-1-(-5))/(9-2))`: it returns -1.75.
4. Check: type the circle `(x-2)^2+(y+5)^2=(9-2)^2+(-1+5)^2` and the line `y=-1.75(x-9)-1`. The line touches the circle only at $(9, -1)$. The slope is $-\frac{7}{4}$.

Answer: $-\frac{7}{4}$

3. HARD · DESMOS

A circle has a radius of 12 meters. What is the area, in square meters, of a sector with a central angle of 150°? (Use $\pi \approx 3.14$.)

(A) 150.8 (B) 172.3

(C) 188.4 ✓ (D) 205.6

1. Area of the whole circle: type `3.14*12^2` : it returns 452.16.
2. Type `150/360` : it returns 0.4167, the fraction of the circle for 150°.
3. Type `3.14*12^2*150/360` : it returns 188.4, so the sector area is 188.4 square meters, choice (C).

Answer: (C)

4. HARD · DESMOS

A circle in the xy-plane is defined by the equation $x^2 + y^2 - 6x + 10y + 18 = 0$. What is the distance, in units, from the center of the circle to the origin?

(A) $\sqrt{34}$ ✓ (B) 34

(C) 4 (D) $2\sqrt{34}$

1. Type the circle: `x^2+y^2-6x+10y+18=0` .
2. Center: Type `6/2` : it returns 3 and Type `-10/2` : it returns -5, so $C = (3, -5)$.
3. Distance from C to the origin: Type `sqrt(3^2+(-5)^2)` : it returns 5.831.
4. Check: Type `sqrt(34)` : it returns 5.831, the same number, so the distance is $\sqrt{34}$, choice (A).

Answer: (A)

5. HARD · DESMOS

A circle in the xy-plane has center $(0, 3)$ and passes through the point $(5, 15)$. Which of the following is an equation of the circle?

(A) $(x-5)^2 + (y-3)^2 = 169$ (B) $x^2 + (y-3)^2 = 13$

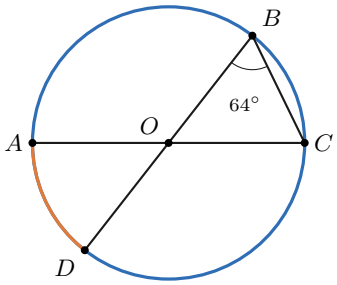
(C) $x^2 + (y-3)^2 = 144$ **(D) $x^2 + (y-3)^2 = 169$ ✓**

1. Type the center `(0,3)` and the point `(5,15)` .
2. The radius squared is the squared distance between them: Type `(5-0)^2+(15-3)^2` : it returns 169.
3. Type the circle `x^2+(y-3)^2=169` : it passes through both points, so it is the correct option, choice (D).
4. The option with 144 on the right fails: $(5, 15)$ is 5 across and 12 up from the center, so Type `5^2+(15-3)^2` : it returns 169, not 144.

Answer: (D)

6. HARD · DESMOS

O is the center of a circle. Chords $\overline{AC}$ and $\overline{BD}$ are diameters that intersect at O (so A, O, C are collinear, and B, O, D are collinear). OB and OC are radii of the circle. If the measure of angle OBC is 64°, what is the measure, in degrees, of arc AD?



1. Triangle OBC is isosceles ($OB = OC$), so $\angle OCB = 64°$. Type `180-2*64`: it returns 52, the angle BOC.

2. $\angle AOD$ is vertical to $\angle BOC$, so it is also 52°, and a central angle equals its arc.

3. Check: Type `(180-52)/2`: it returns 64, which is the 64° given in the base angles.

4. Arc $AD = 52°$.

Answer: 52

7. HARD · DESMOS

A sector of a circle with radius 9 has an area of 18π. The central angle of the sector measures $k\pi$ radians. What is the value of k?

1. Sector area is $\frac{1}{2}r^2\theta$. With $\theta = k\pi$ the area is $\frac{1}{2}(81)k\pi = 18\pi$, so $k = \frac{2 \cdot 18}{81}$: Type `2*18/9^2`: it returns 0.4444.

2. As a fraction: Type `4/9`: it returns 0.4444, so $k = \frac{4}{9}$.

3. Check: type `1/2*9^2*(4/9)*pi`: it returns 56.5487, and Type `18pi`: it returns 56.5487

4. Enter $\frac{4}{9}$ (or 0.444).

Answer: $\frac{4}{9}$

8. HARD · DESMOS

The equation $x^2 + y^2 - 8x + 10y = 8$ defines a circle in the xy-plane. What is the radius of the circle?

1. Type the circle: `x^2+y^2-8x+10y=8`.

2. Center: Type `8/2`: it returns 4 and Type `-10/2`: it returns -5.

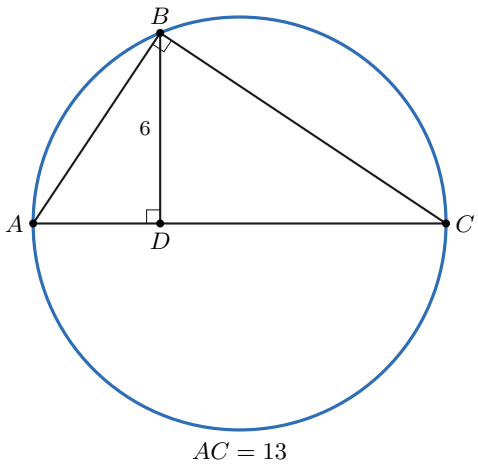
3. Radius: Type `sqrt(8+4^2+5^2)`: it returns 7.

4. Check: the point $(11, -5)$ (seven units right of the center) is on the circle, so the radius is 7.

Answer: 7

9. HARD · DESMOS

Points A, B, and C lie on a circle, and $\angle ABC$ is a right angle. Point D lies on segment AC such that $BD \perp AC$, with $BD = 6$. The diameter of the circle is 13 and $AB < BC$. What is the value of $\dfrac{CD}{AD}$?

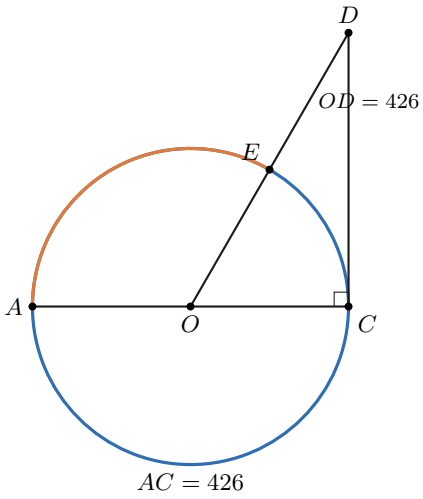


1. $\angle ABC = 90°$ makes $AC = 13$ a diameter, so $AD + CD = 13$, and the altitude gives $AD \cdot CD = BD^2$: Type `6^2` : it returns 36.
2. Type `y=x(13-x)` and `y=36` (x stands for AD).
3. Tap the two intersections: (4, 36) and (9, 36). Since $AB < BC$, D is closer to A: $AD = 4$, $CD = 9$.
4. Type `9/4` : it returns 2.25, so $\dfrac{CD}{AD} = \dfrac{9}{4}$.

Answer: $\dfrac{9}{4}$

10. HARD · DESMOS

In a circle with center O, segment AC is a diameter and $AC = 426$. Point D lies outside the circle with $OD = AC$, and $\overline{CD}$ is tangent to the circle at C. Segment OD intersects the circle at E. The length of minor arc AE is $b\pi$. What is the value of b?



1. $OC = 213$ and $OD = 426$, and $CD \perp OC$, so $\cos\angle COD = \dfrac{213}{426}$. Type `arccos(213/426)*180/pi`: it returns 60 (degrees).
2. E is on OD, so $\angle AOE = 180 - 60$: Type `180-60`: it returns 120.
3. Arc $AE = \dfrac{120}{360} \cdot 2\pi(213)$, so $b =$ Type `120/360*2*213`: it returns 142.
4. $b = 142$.

Answer: 142

11. HARD · DESMOS

Circle A in the xy-plane has equation $x^2 + y^2 - 8x + 6y - 8n = 0$, where n is a constant. Circle B has the same center as circle A, has twice the diameter of circle A, and passes through $(10, -3)$. What is the value of n?

1. Circle A: center Type `8/2`: it returns 4 and Type `-6/2`: it returns -3, so $(4, -3)$.
2. Circle B has radius Type `10-4`: it returns 6 (distance from $(4, -3)$ to $(10, -3)$), and twice the diameter, so A has radius Type `6/2`: it returns 3.
3. Circle A is $(x-4)^2 + (y+3)^2 = 8n + 25$ and $r^2 = 9$, so Type `(9-25)/8`: it returns -2.
4. Check: with $n = -2$ the circle is `x^2+y^2-8x+6y+16=0`; its radius is Type `sqrt(4^2+3^2-16)`: it returns 3.

Answer: -2

12. HARD · DESMOS

In the xy-plane, a circle passes through the points $(-1, 2)$, $(7, 2)$ and $(6, 3)$. Which of the following is an equation of the circle?

(A) $(x-3)^2+(y+1)^2=5$

(B) $(x-3)^2+(y-1)^2=17$

(C) $(x-3)^2+(y+1)^2=25$ ✓

(D) $(x-3)^2+(y-2)^2=16$

1. Plot the three points: type `(-1,2)`, `(7,2)` and `(6,3)`.

2. Try the option $(x-3)^2+(y+1)^2=25$: type `(x-3)^2+(y+1)^2=25`. The circle passes through all three points.

3. Check the third point by hand: Type `(6-3)^2+(3+1)^2`: it returns 25, which equals $r^2=25$. The option $(x-3)^2+(y-2)^2=16$ fails: Type `(6-3)^2+(3-2)^2`: it returns 10, not 16, so its circle misses $(6, 3)$.

Answer: (C)

13. HARD · DESMOS

In the xy-plane, the line $x+y=4$ intersects the circle $x^2+y^2-4x+2y=24$ at two points. What is the distance between these two points?

(A) 7

(B) $7\sqrt{2}$ ✓

(C) 14

(D) $\dfrac{7\sqrt{2}}{2}$

1. Type the circle `x^2+y^2-4x+2y=24` and the line `x+y=4`.

2. Tap the two intersection points: (0, 4) and (7, −3).

3. Type `sqrt((7-0)^2+(-3-4)^2)`: it returns 9.8995, which is the distance between them.

4. Exact value: $\sqrt{49+49}=\sqrt{98}=7\sqrt{2}$, and $7\sqrt{2}\approx 9.8995$, choice (B).

Answer: (B)

14. HARD · DESMOS

In the xy-plane, the line $y=k(x-1)-2$, where k is a positive constant, is tangent to the parabola $y=x^2+2x+4$. What is the value of k?

(A) -2

(B) 4

(C) 8

(D) 10 ✓

1. Type the parabola `y=x^2+2x+4`, then the line `y=k(x-1)-2` and accept the slider for k.

2. Move k until the line just touches the parabola at one point. For a positive k this happens at $k=10$; tap the touching point: (4, 28).

3. Check: Type `10*(4-1)-2`: it returns 28 and Type `4^2+2*4+4`: it returns 28, the same height, so the line meets the parabola at $x=4$.

4. The slider also touches at $k=-2$, but the question says k is positive, so $k=10$, choice (D).

Answer: (D)

15. HARD · DESMOS

$$x^2 + y^2 - 6x + 8y = k$$

In the xy-plane, the graph of the equation above, where k is a constant, is a circle that is tangent to the line $4x - 3y = -11$. What is the value of k?

1. Center of $x^2 + y^2 - 6x + 8y = k$: Type `6/2` : it returns 3 and Type `-8/2` : it returns -4, so $(3, -4)$.
2. Radius = distance from the center to the line $4x - 3y + 11 = 0$: Type `abs(4*3-3*(-4)+11)/sqrt(4^2+3^2)` : it returns 7.
3. $r^2 = k + 3^2 + 4^2$, so $k = r^2 - 25$: Type `7^2-3^2-4^2` : it returns 24.
4. Check: type `x^2+y^2-6x+8y=24` and `4x-3y=-11` ; they touch at one point, $(-2.6, 0.2)$.

Answer: 24

16. HARD · DESMOS

$$(x - a - 2)^2 + (y - 2a + 1)^2 = 25$$

In the xy-plane, the circle above is defined for each real number a. Which of the following points lies on the circle for every real value of a?

(A) $(a + 5,\ 2a + 3)$ ✓

(B) $(a + 5,\ 2a + 4)$

(C) $(2a + 5,\ 2a + 3)$

(D) $(a - 1,\ 2a + 2)$

1. Type the circle `(x-a-2)^2+(y-2a+1)^2=25` and accept the slider for a; the circle slides as a changes.
2. Type the candidate point `(a+5,2a+3)` : it stays on the circle while you move a. Check the distance to the center $(a + 2, 2a - 1)$: Type `((a+5)-(a+2))^2+((2a+3)-(2a-1))^2` : it returns 25 for every value of a, which is r^2.
3. The point `(a+5,2a+4)` leaves the circle: Type `((a+5)-(a+2))^2+((2a+4)-(2a-1))^2` : it returns 34 for every value of a, not 25.
4. The point that always lies on the circle is the one in choice (A).

Answer: (A)

17. HARD · DESMOS

$$(x-1)^2+(y+3)^2=40$$

$$x^2+y^2-4kx+2ky=9$$

In the xy-plane, the two equations above define circles. The center of the second circle lies on the first circle. If k is a positive constant, what is the value of k?

(A) $1+2\sqrt{7}$ **(B) $1+\sqrt{7}$ ✓**

(C) $-1+\sqrt{7}$ (D) $2+\sqrt{7}$

1. Second circle: x^2-4kx and y^2+2ky give center $(2k,-k)$. It must lie on the first circle: $(2k-1)^2+(-k+3)^2=40$, i.e. $k^2-2k-6=0$.
2. Type `y=x^2-2x-6` and tap the zeros: (3.6458, 0) and (−1.6458, 0).
3. Take the positive one: Type `1+sqrt(7)` : it returns 3.6458.
4. $k=1+\sqrt{7}$, choice (B).

Answer: (B)

18. HARD · DESMOS

In the xy-plane, a circle has equation $x^2+y^2+Dx+Ey+F=0$, where D, E, and F are constants. The circle passes through the points $(2,7)$ and $(10,7)$, has radius 5, and has its center below the line $y=7$. What is the value of $E+F$?

1. The points $(2,7)$ and $(10,7)$ share y, so the center's x is Type `(2+10)/2` : it returns 6.
2. Center $(6,m)$, radius 5: $m=7-\sqrt{25-4^2}$ (below $y=7$): Type `7-sqrt(5^2-(6-2)^2)` : it returns 4.
3. $D=-12$, $E=-8$, $F=6^2+4^2-25$: Type `6^2+4^2-25` : it returns 27; type the circle `x^2+y^2-12x-8y+27=0` : it passes through (2, 7) and (10, 7).
4. $E+F=$ Type `-8+27` : it returns 19.

Answer: 19

19. HARD · DESMOS

$$x^2 + y^2 - 4x + 6y - 12 = 0$$

In the xy-plane, the graph of the equation above is a circle. Line ℓ has slope $\frac{3}{4}$ and is tangent to the circle at a point with a positive y-coordinate. What is the y-intercept of ℓ?

(A) $-\frac{43}{4}$ (B) 1

(C) $\frac{7}{4}$ ✓ (D) $-\frac{1}{3}$

1. Type the circle `x^2+y^2-4x+6y-12=0`; its center is $(2, -3)$ and its radius is Type `sqrt(12+2^2+3^2)`: it returns 5.
2. A tangent of slope $\frac{3}{4}$ touches where the radius has slope $-\frac{4}{3}$, i.e. at center $\pm 5 \cdot \frac{(-3, 4)}{5}$. The point with positive y: Type `2-5*3/5`: it returns -1 and Type `-3+5*4/5`: it returns 1, so $(-1, 1)$.
3. y-intercept: Type `1-3/4*(-1)`: it returns 1.75.
4. Type `y=3/4x+7/4`: it touches the circle once, at $(-1, 1)$. The y-intercept is $\frac{7}{4}$, choice (C).

Answer: (C)

20. HARD · DESMOS

In the xy-plane, the circle with equation $(x - 7)^2 + (y + 2)^2 = 9$ has two distinct tangent lines that pass through the point $(2, -2)$. What is the product of the slopes of these two lines?

1. Type the circle `(x-7)^2+(y+2)^2=9` and the point `(2,-2)`: the point is 5 units left of the center, so $\sin\theta = \frac{3}{5}$.
2. Slope of each tangent: Type `tan(arcsin(3/5))`: it returns 0.75 and its negative.
3. Check with a slider: type `y+2=m(x-2)`; at $m = 0.75$ the line touches the circle once (and at $m = -0.75$ as well).
4. Product: Type `(3/4)*(-3/4)`: it returns -0.5625, so $-\frac{9}{16}$.

Answer: $-\frac{9}{16}$

21. HARD · DESMOS

In the xy-plane, a circle lies in the first quadrant and is tangent to both the x-axis and the y-axis. The line $3x + 4y = 20$ intersects the circle at two points 8 units apart. What is the sum of all possible values of the radius of the circle?

1. The circle is tangent to both axes in the first quadrant, so its center is (r, r) and its radius is r. The distance from the center to the line $3x + 4y - 20 = 0$ is $\frac{|3r + 4r - 20|}{5} = \frac{|7r - 20|}{5}$.
2. The chord is 8 long, so half of it is 4, and $r^2 = 4^2 + \left(\frac{7r - 20}{5}\right)^2$. Multiply by 25: $25r^2 = 400 + 49r^2 - 280r + 400$, which simplifies to $3r^2 - 35r + 100 = 0$.
3. Type `y=3x^2-35x+100` and tap the two x-intercepts: (5, 0) and (6.6667, 0).
4. Type `5+20/3`: it returns 11.6667, which is $\frac{35}{3}$ (also $-\frac{-35}{3}$ by the sum of roots).

Answer: $\frac{35}{3}$

22. HARD · DESMOS

In the xy-plane, the center of a circle lies on the y-axis. The circle passes through the points $(6, 1)$ and $(-2, 5)$. Which of the following is an equation of the circle?

(A) $x^2 + (y+1)^2 = 40$ ✓

(B) $x^2 + (y-1)^2 = 40$

(C) $x^2 + (y+1)^2 = 20$

(D) $x^2 + (y+1)^2 = 44$

1. Type the points `(6,1)` and `(-2,5)`. The center is $(0, k)$ with equal distances: $36 + (1-k)^2 = 4 + (5-k)^2$.

2. This simplifies to $37 - 2k = 29 - 10k$, so $k =$ Type `(29-37)/8`: it returns -1.

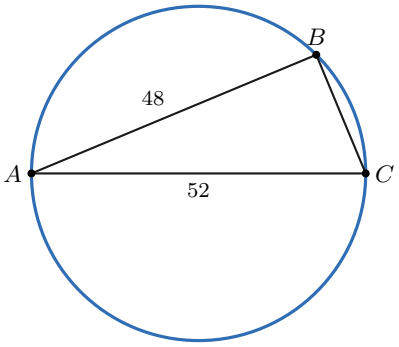
3. $r^2 = 36 + (1-k)^2$: Type `6^2+(1-(-1))^2`: it returns 40.

4. Type `x^2+(y+1)^2=40`: it passes through both points ((6, 1) and $(-2, 5)$). Choice (A).

Answer: (A)

23. HARD · DESMOS

Triangle ABC is inscribed in a circle with circumference 52π. Side AC has length 52 and side AB has length 48. What is the ratio of the length of the shortest side of the triangle to the radius of the circle?



(A) $\frac{24}{13}$

(B) $\frac{5}{12}$

(C) $\frac{5}{13}$

(D) $\frac{10}{13}$ ✓

1. Radius: Type `52/2`: it returns 26, so the diameter is $52 = AC$ and $\angle B = 90°$.

2. BC: Type `sqrt(52^2-48^2)`: it returns 20, the shortest side.

3. Ratio: Type `20/26`: it returns 0.7692.

4. Check: Type `10/13`: it returns 0.7692 is the same number, choice (D).

Answer: (D)

24. HARD · DESMOS

In the xy-plane, the line $y = mx + 10$, where m is a constant, intersects the circle $x^2 + y^2 = 12$ at exactly one point. What is the value of m^2?

1. One common point means the line $y = mx + 10$ is tangent to $x^2 + y^2 = 12$. Type `x^2+y^2=12` and `y=mx+10` with a slider for m.
2. Distance from the center to the line: $\frac{10}{\sqrt{m^2+1}} = \sqrt{12}$, so $m^2 = \frac{100}{12} - 1$: Type `100/12-1`: it returns 7.3333.
3. As a fraction: Type `22/3`: it returns 7.3333. The slider touches at $m = \pm$ Type `sqrt(22/3)`: it returns 2.708.
4. Check: Type `10/sqrt(22/3+1)`: it returns 3.4641 equals Type `sqrt(12)`: it returns 3.4641. So $m^2 = \frac{22}{3}$.

Answer: $\frac{22}{3}$

25. HARD · DESMOS

In the xy-plane, the circle $x^2 + y^2 + 8x - 2y - 8 = 0$ and the line $5x + 12y = 44$ intersect at two points, P and Q. Let C be the center of the circle. What is the area of triangle CPQ?

(A) $\frac{25}{2}$ (B) 6

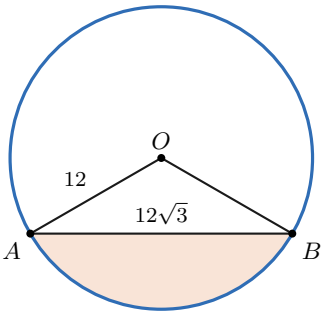
(C) 12 ✓ (D) 24

1. Type `x^2+y^2+8x-2y-8=0` and `5x+12y=44`; tap the two intersections: (0.3077, 3.5385) and (−5.2308, 5.8462).
2. Center $C = (-4, 1)$. Its distance to the line is the height of the triangle: Type `abs(5*(-4)+12*1-44)/sqrt(5^2+12^2)`: it returns 4.
3. Half the chord: Type `sqrt(25-4^2)`: it returns 3, so $PQ = 6$.
4. Area: Type `1/2*6*4`: it returns 12, choice (C).

Answer: (C)

26. Hard · Desmos

In the circle shown, O is the center, the radius is 12, and chord $\overline{AB}$ has length $12\sqrt{3}$. What is the area of the region bounded by chord $\overline{AB}$ and the minor arc AB?



(A) $24\pi - 36\sqrt{3}$

(B) $48\pi - 36\sqrt{3}$ ✓

(C) 48π

(D) $96\pi + 36\sqrt{3}$

1. Angle at the center: Type `arccos((12^2+12^2-(12*sqrt(3))^2)/(2*12*12))*180/pi` : it returns 120 degrees.

2. Sector: Type `120/360*pi*12^2` : it returns 150.7964. Triangle: Type `1/2*12*12*sin(120 deg)` : it returns 62.3538.

3. Segment = sector − triangle: Type `120/360*pi*12^2-1/2*12*12*sin(120 deg)` : it returns 88.4426.

4. Check: Type `48pi-36sqrt(3)` : it returns 88.4426 is the same, choice (B).

Answer: (B)

27. Hard · Desmos

A circular clock face has radius 15 centimeters. A shaded sector of the clock face has area 45π square centimeters. What is the perimeter, in centimeters, of the shaded sector (the arc together with the two bounding radii)?

(A) $15 + 6\pi$

(B) $30 + 3\pi$

(C) $30 + 45\pi$

(D) $30 + 6\pi$ ✓

1. Sector area $\frac{1}{2}r^2\theta = 45\pi$, so $\theta = \frac{2 \cdot 45\pi}{15^2}$: Type `2*45*pi/15^2` : it returns 1.2566 radians.

2. Arc length $r\theta$: Type `15*2*45*pi/15^2` : it returns 18.8496 (that is 6π).

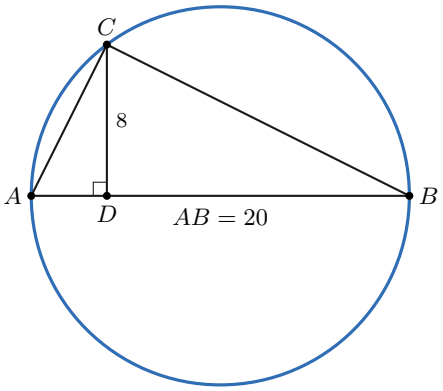
3. Perimeter: Type `2*15+15*2*45*pi/15^2` : it returns 48.8496.

4. Check: Type `30+6pi` : it returns 48.8496, choice (D).

Answer: (D)

28. HARD · DESMOS

Segment $\overline{AB}$ is a diameter of a circle and has length 20. Point C lies on the circle, and point D lies on $\overline{AB}$ with $\overline{CD} \perp \overline{AB}$ and $CD = 8$. If $AD < DB$, what is the length of $\overline{AC}$?



(A) $4\sqrt{5}$ ✓

(B) $8\sqrt{5}$

(C) $4\sqrt{17}$

(D) 10

1. $\angle ACB = 90°$ and CD is the altitude, so $AD \cdot DB = CD^2$: Type `8^2` : it returns 64, with $AD + DB = 20$.

2. Type `y=x(20-x)` and `y=64` ; tap the intersections: (4, 64) and (16, 64). $AD < DB$, so $AD = 4$.

3. $AC = \sqrt{AD^2 + CD^2}$: Type `sqrt(4^2+8^2)` : it returns 8.9443.

4. Check: Type `4sqrt(5)` : it returns 8.9443, choice (A).

Answer: (A)

29. HARD · DESMOS

A circle has area 48π. A square and an equilateral triangle are each inscribed in the circle (all vertices on the circle). By how much does the area of the square exceed the area of the triangle?

(A) $96 - 27\sqrt{3}$

(B) $96 - 36\sqrt{3}$ ✓

(C) $96 - 12\sqrt{3}$

(D) $192 - 36\sqrt{3}$

1. $r^2 = 48$ (area 48π). The square has diagonal $2r$, so its area is Type `2*48` : it returns 96.

2. Equilateral triangle: area $\frac{3\sqrt{3}}{4}r^2$: Type `3*sqrt(3)/4*48` : it returns 62.3538.

3. Difference: Type `96-3*sqrt(3)/4*48` : it returns 33.6462.

4. Check: Type `96-36sqrt(3)` : it returns 33.6462, choice (B).

Answer: (B)

30. HARD · DESMOS

In the xy-plane, the circle $x^2 + y^2 + 8x - 2y = 8$ and the line $5x - 12y = k$, where k is a constant, intersect at exactly one point. What is the greatest possible value of k?

(A) 65 (B) 32

(C) 33 ✓ (D) -97

1. Type the circle `x^2+y^2+8x-2y=8`: center $(-4, 1)$, radius Type `sqrt(8+4^2+1^2)`: it returns 5.

2. Tangent means distance $= 5$: $\dfrac{|k+32|}{13} = 5$. Type `y=abs(x+32)/13` and `y=5` and tap: (33, 5) and $(-97, 5)$.

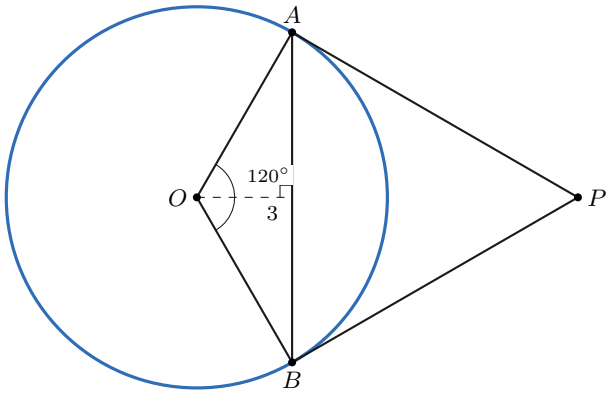
3. The greatest k is 33. Check: type `5x-12y=33`; it touches the circle once. Type `abs(5*(-4)-12*1-33)/13`: it returns 5

4. $k = 33$, choice (C).

Answer: (C)

31. HARD · DESMOS

In a circle with center O, chord $\overline{AB}$ has $\angle AOB = 120°$ and lies at a perpendicular distance of 3 from O. The lines tangent to the circle at A and at B intersect at point P. What is the length of $\overline{PA}$?



(A) $6\sqrt{3}$ ✓ (B) 12

(C) $3\sqrt{3}$ (D) 6

1. Half of $\angle AOB$ is Type `120/2`: it returns 60 degrees; $\cos 60° = \dfrac{3}{r}$, so $r =$ Type `3/cos(60 deg)`: it returns 6.

2. $PA \perp OA$ and $\angle AOP = 60°$, so $PA = r \tan 60°$: Type `6*tan(60 deg)`: it returns 10.3923.

3. Check: Type `6sqrt(3)`: it returns 10.3923, choice (A).

Answer: (A)

32. HARD · DESMOS

In the xy-plane, a circle is tangent to both the x-axis and the y-axis, has its center in the second quadrant, and passes through the point $(-2, 9)$. The diameter of the circle is less than 15. The circle has equation $x^2+y^2+Dx+Ey+F = 0$, where D, E and F are constants. What is the value of F?

1. Center $(-r, r)$, radius r, through $(-2, 9)$: $(r-2)^2 + (9-r)^2 = r^2$. Type `y=x^2-22x+85` and tap the zeros: (5, 0) and (17, 0).

2. Diameter under 15: Type `2*5`: it returns 10 works, while $2 \cdot 17 = 34$ does not, so $r = 5$.

3. $F = h^2 + k^2 - r^2$: Type `(-5)^2+5^2-5^2`: it returns 25.

4. Check: Type `(-2+5)^2+(9-5)^2`: it returns 25 equals $r^2 = 25$, so $(-2, 9)$ is on $(x+5)^2 + (y-5)^2 = 25$.

Answer: 25

33. HARD · DESMOS

$$x^2 + y^2 + 10x - 24y = 0$$

In the xy-plane, the graph of the equation above is a circle. What is the diameter of the circle?

1. Type the circle `x^2+y^2+10x-24y=0` (it passes through the origin).

2. Center: Type `-10/2`: it returns -5 and Type `24/2`: it returns 12.

3. Radius: Type `sqrt(0+5^2+12^2)`: it returns 13.

4. Diameter: Type `2*13`: it returns 26.

Answer: 26

34. HARD · DESMOS

In the xy-plane, a circle has its center on the line $y = 2x$ and passes through the points $(4, 14)$ and $(-1, -1)$. What is the square of the radius of the circle?

1. Type the points `(4,14)` and `(-1,-1)` and the line `y=2x` (the center lies on it).

2. The center is on the perpendicular bisector of the two points (midpoint $(1.5, 6.5)$, slope $-\frac{1}{3}$). Type `y=-1/3(x-3/2)+13/2`.

3. Tap the intersection with the line: (3, 6), the center.

4. r^2: Type `(3-4)^2+(6-14)^2`: it returns 65.

Answer: 65

35. HARD · DESMOS

In the xy-plane, the circle $(x-2)^2+(y+1)^2=25$ passes through the point $(6,2)$. The line tangent to the circle at $(6,2)$ crosses the x-axis at the point $(a,0)$. What is the value of a?

(A) $\frac{26}{3}$ (B) $\frac{3}{2}$

(C) $\frac{10}{3}$ **(D) $\frac{15}{2}$ ✓**

1. Type the circle `(x-2)^2+(y+1)^2=25` and the point `(6,2)` .

2. Radius slope Type `(2+1)/(6-2)` : it returns 0.75; tangent slope Type `-1/((2+1)/(6-2))` : it returns −1.3333.

3. Type the tangent `y=-4/3(x-6)+2` and tap its x-intercept: (7.5, 0).

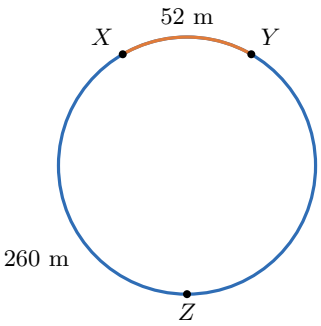
4. Check: Type `6+(0-2)/(-4/3)` : it returns 7.5. $a=\frac{15}{2}$, choice (D).

Answer: (D)

PART B · SELF-ASSESSMENT — 21 PROBLEMS

36. MEDIUM · DESMOS

Points X, Y, and Z lie on a circular fountain, as shown below. On this circle, minor arc XY has a length of 52 meters, and major arc XZY has a length of 260 meters. What is the circumference, in meters, of the circle?



(A) 208

(B) 260

(C) 312 ✓

(D) 52

1. The minor and major arcs together go once around the circle, so the circumference is their sum. Type `52+260`: it returns 312.
2. Type `52/312`: it returns 0.1667, so the 52 m arc is $\frac{1}{6}$ of the circle, which fits a short minor arc.
3. The circumference is 312 meters, choice (C).

Answer: (C)

37. MEDIUM · DESMOS

What is the center of the circle in the xy-plane defined by the equation $x^2 + y^2 - 14x + 22y + 10 = 0$?

(A) $(-7, -11)$

(B) $(7, -11)$ ✓

(C) $(7, 11)$

(D) $(-7, 11)$

1. Type the circle: `x^2+y^2-14x+22y+10=0`.
2. The center's coordinates are $-\frac{D}{2}$ and $-\frac{E}{2}$: Type `14/2`: it returns 7 and Type `-22/2`: it returns -11.
3. Check: type `(x-7)^2+(y+11)^2=7^2+11^2-10` (center $(7, -11)$, $r^2 = 7^2 + 11^2 - 10$). It lies exactly on top of the first circle.
4. The center is $(7, -11)$, choice (B).

Answer: (B)

38. Medium · Desmos

A circle in the xy-plane has equation $(x+2)^2+(y-5)^2=49$. What is the radius of the circle?

(A) 14

(B) 24.5

(C) 49

(D) 7 ✓

1. Type the circle: `(x+2)^2+(y-5)^2=49`.
2. Draw the horizontal line through the center, `y=5`, and tap the two ends: $(-9, 5)$ and $(5, 5)$.
3. Type `(5-(-9))/2`: it returns 7, which is the radius.
4. Check: Type `sqrt(49)`: it returns 7. The radius is 7, choice (D).

Answer: (D)

39. Medium · Desmos

A circle has a radius of 9 cm. What is the length, in cm, of an arc that subtends a central angle of 80°? (Use $\pi \approx 3.14$.)

(A) 12.6 ✓

(B) 15.7

(C) 18.8

(D) 9.4

1. Circumference: type `2*3.14*9`: it returns 56.52.
2. Type `80/360`: it returns 0.2222, the fraction of the circle for 80°.
3. Type `2*3.14*9*80/360`: it returns 12.56, so the arc is about 12.6 cm, choice (A).

Answer: (A)

40. Medium · Desmos

The equation of a circle in the xy-plane is $x\,(x-8)+y\,(y+6)=39$. What is the radius of the circle?

(A) 7

(B) 8 ✓

(C) 9

(D) 6

1. Type the equation exactly as given: `x(x-8)+y(y+6)=39`. Desmos draws the circle.
2. Expanded, it is $x^2-8x+y^2+6y=39$, so the center is Type `8/2`: it returns 4 and Type `-6/2`: it returns -3.
3. Radius: Type `sqrt(39+4^2+3^2)`: it returns 8.
4. Check: draw `y=-3` and tap its ends $(-4, -3)$ and $(12, -3)$; Type `(12-(-4))/2`: it returns 8. The radius is 8, choice (B).

Answer: (B)

41. Medium · Desmos

A circle has an area of 144π ft^2. What is the circumference of the circle?

(A) 12 ft

(B) 12π ft

(C) 24π ft ✓

(D) 48π ft

1. The area is $\pi r^2 = 144\pi$, so $r =$ Type `sqrt(144)` : it returns 12.

2. Circumference: Type `2*pi*12` : it returns 75.3982.

3. Check against 24π: Type `24*pi` : it returns 75.3982. The circumference is 24π ft, choice (C).

Answer: (C)

42. Medium · Desmos

A circle has radius $2x - 3$ and circumference $(3x + 4)\pi$. What is the area of the circle?

(A) 289π ✓

(B) 17π

(C) 34π

(D) 196π

1. Circumference $2\pi r$ equals $(3x + 4)\pi$, so $2(2x - 3) = 3x + 4$. Type `y=2(2x-3)` and `y=3x+4` .

2. Tap the intersection: (10, 34), so $x = 10$.

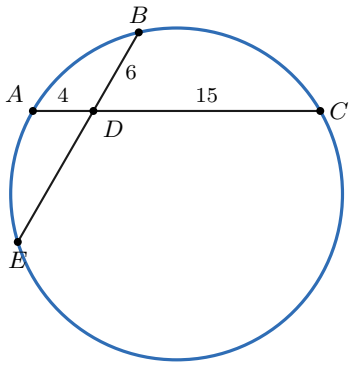
3. Radius: Type `2*10-3` : it returns 17.

4. Area: Type `pi*17^2` : it returns 907.9203, and Type `289pi` : it returns 907.9203 is the same value, choice (A).

Answer: (A)

43. Medium · Desmos

Chords $\overline{AC}$ and $\overline{BE}$ of a circle intersect at point D inside the circle. If $AD = 4$, $DC = 15$, and $BD = 6$, what is the length of $\overline{BE}$?



1. Crossing chords: $AD \cdot DC = BD \cdot DE$, that is $4 \cdot 15 = 6 \cdot DE$. Type `y=6x` and `y=4*15` .

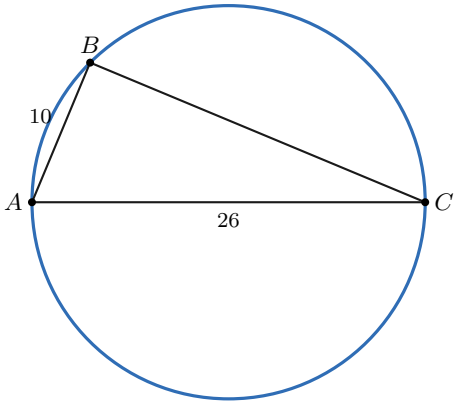
2. Tap the intersection: (10, 60), so $DE = 10$.

3. Type `6+10` : it returns 16, the length $BE = BD + DE$.

Answer: 16

44. Medium · Desmos

Segment AC is a diameter of a circle, and point B lies on the circle. If $AC = 26$ and $AB = 10$, what is the length of BC?



(A) $\sqrt{776}$

(B) 26

(C) 16

(D) 24 ✓

1. AC is a diameter, so $\angle ABC = 90°$ and AC is the hypotenuse.
2. Type `sqrt(26^2-10^2)` : it returns 24.
3. The wrong subtraction $26 - 10$ would give Type `26-10` : it returns 16; adding squares gives Type `sqrt(26^2+10^2)` : it returns 27.8568, which is not a choice.
4. $BC = 24$, choice (D).

Answer: (D)

45. Medium · Desmos

The equation $x^2 - 18x + y^2 - 6y = -38$ represents a circle in the xy-plane. What is the area of the circle?

(A) $2\sqrt{13}\,\pi$

(B) 43π

(C) 52π ✓

(D) 128π

1. Type the circle: `x^2-18x+y^2-6y=-38` .
2. Center: Type `18/2` : it returns 9 and Type `6/2` : it returns 3.
3. $r^2 = -38 + 9^2 + 3^2$: Type `-38+9^2+3^2` : it returns 52.
4. Area $= \pi r^2$: Type `pi*52` : it returns 163.3628, which equals 52π, choice (C).

Answer: (C)

46. Medium · Desmos

A circle has radius 15. An arc of the circle has length 6π. What is the measure, in degrees, of the central angle that subtends the arc?

(A) 18 **(B) 72 ✓**

(C) 36 (D) 144

1. Circumference: Type `2*pi*15` : it returns 94.2478 (that is 30π).

2. Fraction of the circle: Type `6*pi/(2*pi*15)` : it returns 0.2.

3. Central angle: Type `6*pi/(2*pi*15)*360` : it returns 72 degrees, choice (B).

Answer: (B)

47. Medium · Desmos

A sector of a circle with radius 6 has area 12π. What is the measure, in degrees, of the central angle of the sector?

1. Area of the circle: Type `pi*6^2` : it returns 113.0973.

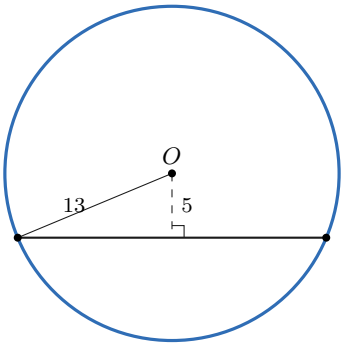
2. Fraction: Type `12*pi/(pi*6^2)` : it returns 0.3333.

3. Angle: Type `12/36*360` : it returns 120 degrees.

Answer: 120

48. Medium · Desmos

A chord of a circle with radius 13 is 5 units from the center of the circle. What is the length of the chord?



1. Type the circle `x^2+y^2=13^2` and the line `y=-5` (the chord is 5 units from the center).

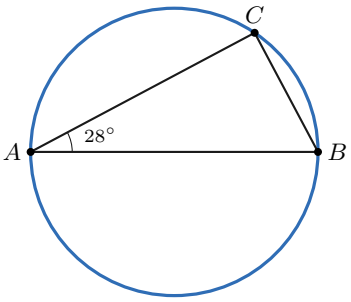
2. Tap the ends of the chord: $(-12, -5)$ and $(12, -5)$.

3. Length: Type `12-(-12)` : it returns 24.

Answer: 24

49. Medium · Desmos

Points A and B are the endpoints of a diameter of a circle, and point C is on the circle. The measure of angle CAB is 28°. What is the measure, in degrees, of angle ABC?



1. AB is a diameter, so $\angle ACB = 90°$.

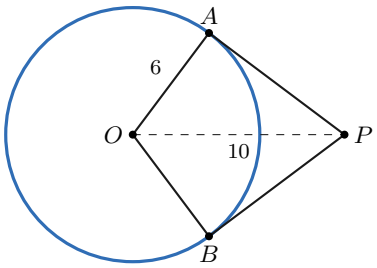
2. Type `180-90-28` : it returns 62.

3. Check: Type `28+62+90` : it returns 180, so $\angle ABC = 62°$.

Answer: 62

50. Hard · Desmos

Point P is 10 units from the center O of a circle with radius 6. Segments $\overline{PA}$ and $\overline{PB}$ are tangent to the circle at A and B. What is the area of quadrilateral $PAOB$?



1. $\angle PAO = 90°$, so $PA =$ Type `sqrt(10^2-6^2)` : it returns 8.

2. Triangle PAO: Type `1/2*8*6` : it returns 24.

3. Two congruent triangles: Type `2*(1/2*8*6)` : it returns 48.

Answer: 48

51. Medium · Desmos

In the xy-plane, a circle has center $(-3, 5)$ and is tangent to the x-axis. Which of the following is an equation of the circle?

(A) $(x+3)^2 + (y-5)^2 = 9$

(B) $(x+3)^2 + (y-5)^2 = 5$

(C) $(x-3)^2 + (y+5)^2 = 25$

(D) $(x+3)^2 + (y-5)^2 = 25$ ✓

1. Type the center `(-3,5)` and think of the x-axis `y=0`.

2. The radius is the distance to the axis, 5. Type `(x+3)^2+(y-5)^2=25`: it touches the x-axis once, at $(-3, 0)$.

3. $r^2 =$ Type `5^2`: it returns 25, so the equation is the one in choice (D).

Answer: (D)

52. Medium · Desmos

In the xy-plane, the line $y = 6$ intersects the circle $x^2 + y^2 = 100$ at two points. What is the distance between the two points?

1. Type `x^2+y^2=100` and `y=6`.

2. Tap the two points: $(-8, 6)$ and $(8, 6)$.

3. Distance: Type `8-(-8)`: it returns 16.

Answer: 16

53. Medium · Desmos

An arc of a circle has length 10π centimeters and subtends a central angle of $\dfrac{2\pi}{3}$ radians. What is the radius of the circle, in centimeters?

1. Arc length $= r\theta$, so $r = \dfrac{10\pi}{2\pi/3}$. Type `10pi/(2pi/3)`: it returns 15.

2. Check: Type `15*(2*pi/3)`: it returns 31.4159 equals Type `10pi`: it returns 31.4159.

3. The radius is 15 centimeters.

Answer: 15

54. Medium · Desmos

Two concentric circles have the same center. The outer circle has radius 7, and the area of the region between the two circles is 48π. What is the radius of the inner circle?

1. Ring area: $\pi \cdot 7^2 - \pi r^2 = 48\pi$, so $r^2 = 7^2 - 48$: Type `7^2-48`: it returns 1.

2. Radius: Type `sqrt(1)`: it returns 1.

3. Check: Type `pi*7^2-pi*1^2`: it returns 150.7964 equals Type `48pi`: it returns 150.7964.

Answer: 1

55. Medium · Desmos

A sector of a circle has radius 12 and perimeter $24 + 4\pi$. What is the measure, in degrees, of the central angle of the sector?

(A) 60 ✓ (B) 120

(C) 240 (D) 30

1. Perimeter minus the two radii gives the arc: Type `24+4pi-2*12` : it returns 12.5664 (that is 4π).

2. Angle $= \frac{\text{arc}}{2\pi r} \cdot 360$: Type `(24+4pi-2*12)/(2pi*12)*360` : it returns 60.

3. Check: Type `60/360*2pi*12` : it returns 12.5664 matches the arc, choice (A).

Answer: (A)

56. Hard · Desmos

A circle has area 50π. A square is inscribed in the circle so that all four vertices lie on the circle. What is the area of the square?

1. Radius: Type `sqrt(50)` : it returns 7.0711. The square's diagonal is the diameter: Type `2*sqrt(50)` : it returns 14.1421.

2. Square area $= \frac{d^2}{2}$: Type `(2*sqrt(50))^2/2` : it returns 100.

3. Check: the circle's area Type `pi*50` : it returns 157.0796 is bigger than the square: the square is 100.

Answer: 100